

Graph Neural Networks and Graph Foundation Models for Scientific Discovery: A Comprehensive Survey

Abstract

Graph-structured data provides a mathematically rigorous abstraction for complex relational systems across the natural and social sciences, ranging from atomic interactions in molecular structures to macroscopic dependencies in biomedical knowledge graphs and physical simulations. Over the past decade, deep graph learning, primarily driven by Graph Neural Networks (GNNs), has achieved remarkable empirical success in accelerating scientific discovery by acting as rapid surrogate models for computationally expensive first-principles calculations. However, traditional GNNs are inherently constrained by localized message-passing mechanisms, limiting their theoretical expressive power and rendering them susceptible to topological pathologies such as over-smoothing and over-squashing. More importantly, the traditional end-to-end training paradigm produces fragmented, task-specific models that fail to generalize to out-of-distribution topological structures. Motivated by the transformative capabilities of foundation models in natural language processing and computer vision, the paradigm is rapidly shifting towards Graph Foundation Models (GFMs). GFMs aim to acquire broad, transferable topological and semantic invariants from massive-scale, multi-domain graph corpora through self-supervised pre-training, enabling generalized zero-shot and few-shot inference across diverse downstream tasks. This comprehensive survey systematically examines the transition from specialized GNNs to universal GFMs in the context of scientific discovery. The structural and representational foundations of GNNs are critically reviewed, emphasizing the development of geometric and equivariant architectures designed to strictly preserve continuous Euclidean symmetries in physical systems. The survey then details the architectural innovations necessary for scaling, particularly the development of Graph Transformers and the decoupling of local and global attention mechanisms. Subsequently, the methodology of GFMs is explored, outlining pre-training strategies, the theoretical construction of unified graph vocabularies via graphons, and taxonomic classifications based on generalization scope. The profound impact of these models is illustrated through methodological applications in materials science, zero-shot drug repurposing via massive medical knowledge graphs, and the resolution of complex differential equations through physics-informed networks. Finally, the survey synthesizes the critical challenges impeding the development of universal GFMs, including structural heterogeneity, negative transfer, multimodal alignment, and the establishment of robust neural scaling laws for non-Euclidean data.

Introduction

The acceleration of scientific discovery relies heavily on the ability to accurately model, simulate, and predict the behavior of complex systems. Historically, these predictive tasks were

dominated by computationally expensive, first-principles physical frameworks, such as Density Functional Theory (DFT) in quantum chemistry, molecular dynamics (MD) simulations in structural biology, and finite element analysis in fluid dynamics [77, 88]. While highly accurate, the cubic or exponential scaling complexity of these classical methods severely restricts their applicability to small systems and short temporal horizons. The advent of Artificial Intelligence for Science (AI4Science) has introduced data-driven surrogate models capable of approximating these physical laws with unprecedented computational efficiency [16, 78].

At the core of this scientific AI revolution is the recognition that the fundamental entities of scientific inquiry—atoms within a crystal lattice, residues interacting within a folded protein, celestial bodies in a gravitational system, and diseases linked in a biological taxonomy—are not isolated instances. Rather, they are interconnected nodes existing within continuous, non-Euclidean relational spaces. Standard deep learning architectures, such as Convolutional Neural Networks (CNNs) designed for regular grids and Recurrent Neural Networks (RNNs) designed for sequential data, fail to capture the permutation invariance and complex topological dependencies inherent in scientific data [6]. Consequently, graph representation learning has emerged as the most natural and theoretically sound paradigm for scientific modeling [1, 7].

Graph Neural Networks (GNNs) operate by iteratively aggregating feature information from local topological neighborhoods, effectively capturing the structural dependencies inherent in relational data. Early GNN architectures laid the groundwork for encoding node, edge, and global graph attributes by utilizing localized spectral filters and attention mechanisms [3, 4]. However, the direct application of standard GNNs to physical systems required a fundamental evolution: the incorporation of geometric and physical symmetries. To ensure that molecular property predictions and interatomic force fields remain consistent regardless of how a system is rotated, translated, or reflected in three-dimensional space, researchers developed equivariant GNNs [18, 19, 20]. These networks explicitly preserve continuous Euclidean symmetries, leading to highly data-efficient, physically faithful simulations that have redefined the limits of machine learning potentials [19, 20, 22].

Despite these remarkable advancements, standard deep graph learning has traditionally followed an end-to-end training paradigm. Models are typically initialized randomly and trained from scratch on heavily annotated datasets tailored to extremely narrow tasks [39, 40]. This bespoke approach results in fragmented model development, extreme susceptibility to out-of-distribution structural shifts, and a profound inability to generalize [49, 50]. Furthermore, standard GNNs are fundamentally constrained by their localized message-passing mechanisms, which lead to severe topological pathologies—such as over-smoothing and over-squashing—when attempting to capture the long-range dependencies crucial for understanding complex scientific phenomena [9, 10, 31].

The extraordinary success of large language models (LLMs) and vision transformers has unequivocally demonstrated that models pre-trained on vast amounts of diverse, unlabeled data can develop emergent reasoning capabilities and seamlessly adapt to unseen tasks [39, 51]. Transporting this foundation model paradigm to the graph domain has birthed the concept of Graph Foundation Models (GFMs) [51]. GFMs seek to unify disparate graph topologies, feature dimensions, and task objectives under a single, highly scalable architecture [49, 52]. By leveraging advanced pre-training strategies—such as masked graph autoencoding and graph

contrastive learning—GFM learn transferable topological and semantic invariants that can be fine-tuned, prompted, or queried in a zero-shot manner across entirely novel scientific domains [42, 43].

This survey systematically explores the theoretical foundations, architectural evolutions, and scientific applications of GNNs and GFMs. The manuscript is organized as follows: The Methods section is subdivided into coherent research directions, beginning with the representational foundations of MPNNs, transitioning to geometric equivariance and Graph Transformers, and culminating in the methodological design of GFMs and their domain-specific scientific applications. The Challenges and Prospects section analyzes the theoretical limits of topological transferability, multimodal integration, and neural scaling laws. Finally, the Conclusion synthesizes the broader implications of these technologies for the future of automated scientific discovery.

Methods

The architectural and methodological landscape of graph learning for scientific discovery can be organized into distinct evolutionary phases. This section systematically details the foundational mechanisms of GNNs, the incorporation of physical symmetries via equivariant networks, the resolution of information bottlenecks through Graph Transformers, and the ultimate synthesis of these techniques into scalable Graph Foundation Models (GFMs). Methodological applications demonstrating the utility of these architectures in materials science, biomedicine, and physical dynamics are integrated throughout the subsequent subsections.

Representational Foundations of Graph Neural Networks

The theoretical backbone of modern deep graph learning is the Message Passing Neural Network (MPNN) framework [1, 7]. Let a scientific system be defined as an undirected or directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes (e.g., atoms, proteins) and $\mathcal{E}$ is the set of edges (e.g., chemical bonds, biological interactions). In an MPNN, each node $v \in \mathcal{V}$ is associated with an initial feature vector $\mathbf{h}_v^{(0)}$. The network updates these representations over L discrete layers through a localized aggregation of neighborhood features [3, 4]. The generalized state transition at layer l is formalized as:

$$\begin{aligned}\mathbf{m}_v^{(l)} &= \text{AGGREGATE}^{(l)} \left(\left\{ \text{MSG}^{(l)} \left(\mathbf{h}_v^{(l-1)}, \mathbf{h}_u^{(l-1)}, \mathbf{e}_{uv} \right) : u \in \mathcal{N}(v) \right\} \right) \\ \mathbf{h}_v^{(l)} &= \text{UPDATE}^{(l)} \left(\mathbf{h}_v^{(l-1)}, \mathbf{m}_v^{(l)} \right)\end{aligned}$$

where $\mathcal{N}(v)$ denotes the immediate neighbors of node v , $\mathbf{e}_{uv}$ represents the edge features between u and v , and the functions MSG and UPDATE are parameterized by differentiable multi-layer perceptrons (MLPs). The AGGREGATE function must be

permutation invariant (e.g., sum, mean, or max) to ensure the network is agnostic to the ordering of a node's neighbors [2, 5]. This recursive message-passing mechanism allows a node to gather information from an expanding receptive field, such that an L -layer MPNN captures structural dependencies within an L -hop topological radius [1, 2]. While empirically successful, standard MPNNs suffer from strictly defined theoretical limits regarding their expressive power. Extensive theoretical research has demonstrated that standard message-passing GNNs are, at most, as powerful as the 1-dimensional Weisfeiler-Lehman (1-WL) graph isomorphism test [2, 15]. Consequently, standard GNNs cannot distinguish between certain non-isomorphic regular graphs; they fundamentally fail to detect specific closed-loop structures, substructures, or long-range cycles that are essential for characterizing phenomena such as molecular rings or multi-hop metabolic pathways [2, 15]. Furthermore, the strictly localized nature of message passing introduces severe topological pathologies when applied to large, highly connected, or highly constrained scientific networks. The phenomenon of *over-smoothing* occurs as network depth increases; the repeated application of Laplacian smoothing causes the representations of all nodes in a connected component to asymptotically converge to an indistinguishable mean vector, completely destroying local feature variance [9, 10]. Conversely, *over-squashing* arises in graphs characterized by structural bottlenecks, such as tree-like molecular graphs or citation networks with high negative curvature [9]. As the receptive field grows exponentially with network depth, the network attempts to compress an exponentially increasing volume of distant neighborhood information into a fixed-dimensional node vector. This results in a catastrophic loss of information, preventing the model from capturing long-range atomic or biological interactions [9, 10]. These fundamental representational bottlenecks necessitated the development of novel geometric constraints and attention-based architectures to handle the rigorous computational demands of physical simulation.

Geometric and Equivariant Architectures for Physical Systems

In domains such as computational chemistry, structural biology, and materials science, data predominantly exists as 3D point clouds or geometric graphs where nodes possess explicit spatial coordinates $\mathbf{X} \in \mathbb{R}^{N \times 3}$. A fundamental requirement for modeling these physical systems is strict adherence to Euclidean symmetries. The potential energy of a molecule, for example, is a scalar quantity that must remain invariant regardless of how the overall molecule is rotated, translated, or reflected in space. Conversely, atomic forces and tensorial properties are vector quantities that must rotate equivariantly with the molecule [18, 19, 21]. Standard GNNs, which process arbitrary node features without spatial awareness, do not intrinsically respect these symmetries. If a molecule's Cartesian coordinates are rotated and fed into a standard GNN, the network will likely predict a different energy state, violating fundamental thermodynamic laws [18, 22]. To resolve this, Geometric and Equivariant Graph Neural Networks (EGNNs) encode inductive biases that strictly enforce mathematical transformations under the Special Euclidean group SE(3) (translations and rotations) or the full Euclidean group E(3) (which includes spatial inversions and reflections) [18, 24].

Mechanisms of Equivariance: Scalarization vs. Irreducible Representations

The architectural design of equivariant networks is broadly delineated into two distinct methodologies: scalarization (invariant networks) and irreducible representation (irrep) tensor networks. Scalarization approaches project 3D coordinates into invariant geometric scalars, predominantly utilizing interatomic distances and angles [16, 17]. These distances are then used to modulate invariant node features within the message-passing framework. Architectures such as SchNet and traditional EGNNs pioneered this approach, offering computationally lightweight mechanisms that are strictly invariant to Euclidean transformations [16, 18]. However, scalar networks struggle to accurately resolve complex angular dependencies and many-body interactions necessary for modeling highly constrained atomic environments, as mapping 3D geometry solely to pairwise distances involves an inherent loss of directional information [18, 22].

To capture directional information explicitly, advanced EGNNs employ tensor field networks and spherical harmonics [23, 24]. These models assign features to different rotational orders (e.g., scalars $l = 0$, vectors $l = 1$, tensors $l = 2$) that transform according to the Wigner-D matrices under $SO(3)$ rotations [19, 21]. By utilizing Clebsch-Gordan tensor products, these networks ensure that messages passed between atoms rigorously preserve higher-order geometric information [19, 23]. Notable architectures utilizing this approach include NequIP and SE(3)-Transformers, which achieved state-of-the-art accuracy on molecular force fields by eliminating the need for hand-crafted symmetry functions, though at the cost of high computational overhead due to the complex tensor algebra required during the forward pass [19, 24].

Higher-Order Message Passing and the MACE Architecture

Despite the remarkable accuracy of irrep-based EGNNs, their computational complexity poses a significant barrier to large-scale Molecular Dynamics (MD) simulations. A primary bottleneck is that standard EGNNs rely on 2-body message passing, meaning the message from node j to node i depends solely on the state of those two atoms [20, 21]. To capture complex many-body quantum effects, these networks must stack numerous layers (typically 5 to 6), exacerbating the computational cost and leading to the aforementioned over-smoothing of features.

The MACE (Multi-BACE) architecture introduced a paradigm shift by implementing higher-order equivariant message passing [20]. MACE constructs messages that simultaneously capture interactions among multiple atoms (e.g., 3-body or 4-body correlations) within a single layer. This is achieved by taking the tensor product of the atomic environment with itself multiple times, utilizing symmetric contractions to form high-rank Cartesian tensor products without explicitly summing over all possible triplets or quadruplets [20]. By leveraging 4-body messages, MACE dramatically increases the expressivity of a single message-passing iteration, reducing the required network depth to just two layers. This shallow but highly expressive architecture results in a highly parallelizable model that reaches or exceeds the accuracy of deeper networks on benchmarks like rMD17, while executing significantly faster [20].

Recent advancements have further optimized these geometric capabilities for scalable

deployment. The introduction of Modular Angular-Radial Attention (MARA) integrates spherical attention into SE(3)-equivariant frameworks, decoupling radial and angular dependencies to allow flexible, geometry-aware weighting of local environments without rigid architectural constraints. Furthermore, aggressive quantization strategies—such as 8-bit quantized SO(3) GNNs utilizing magnitude-direction decoupled quantization—have enabled the deployment of these complex equivariant models on resource-constrained edge devices with minimal loss of physical symmetry or predictive accuracy.

Architecture Class	Key Mechanism	Symmetry & Expressivity	Representative Models
Standard MPNN	Localized feature aggregation	Permutation Invariant (1-WL limit)	GCN [3], GAT [4], GIN [2]
Scalar EGNN	Invariant geometric scalars (distances)	E(n) Invariant / Equivariant updates	SchNet [16], EGNN [18]
Irrep Tensor GNN	Spherical harmonics, Clebsch-Gordan	SE(3) / E(3) Equivariant	TFN [23], NequIP [19]
Higher-Order EGNN	Many-body messages, symmetric contraction	SE(3) Equivariant, > 2 -body interactions	MACE [20]

Graph Transformers and the Resolution of Topological Bottlenecks

While EGNNs solved the symmetry problem for constrained physical systems, the topological limitations of local message passing (over-squashing and restricted WL-expressivity) remained unaddressed for general relational graphs. To bypass the rigid local connectivity of MPNNs, researchers adapted the Transformer architecture—originally designed for sequence processing in natural language—to the graph domain [31, 32].

Graph Transformers (GTs) treat nodes as tokens and apply a global, fully-connected self-attention mechanism, allowing every node to attend directly to every other node in the

graph, regardless of their shortest-path distance [31, 34]. This $O(N^2)$ global receptive field completely circumvents over-squashing by providing a direct, unhindered gradient path between any two nodes. However, standard self-attention is inherently permutation-invariant; it possesses no inherent understanding of graph topology. If applied naively, a standard Transformer views a graph merely as an unordered set of isolated nodes, completely ignoring

the structural priors provided by the edges [31, 33].

To inject structural awareness into the global attention mechanism, GTs require sophisticated Positional Encodings (PE) and Structural Encodings (SE) [32, 34]. Positional Encodings capture the absolute or relative position of a node within the global graph geometry. The most robust methods utilize the spectral decomposition of the graph Laplacian matrix [34]. The non-trivial eigenvectors of the Laplacian serve as a generalized coordinate system for the graph, providing a unique spatial signature for each node that is invariant to node indexing [34]. Conversely, Structural Encodings capture local topological patterns, such as node degrees, clustering coefficients, and the presence of specific motifs. Random Walk Structural Encodings (RWSE), which compute the return probability of random walks of varying lengths originating at a node, are highly effective at characterizing a node's local structural role [32].

The integration of MPNNs and Transformers culminated in the General, Powerful, Scalable (GPS) Graph Transformer framework (GraphGPS) [32]. GraphGPS provides a modular blueprint that explicitly decouples local topological aggregation from global attention. In a single GraphGPS layer, a node's representation is processed via two parallel tracks: a standard local MPNN (e.g., GINE) that processes real edge connections, inherently preserving the 1-WL expressivity and sparse structural inductive biases, and a global attention mechanism enriched with dense PE/SE features to resolve long-range dependencies [32]. Crucially, the decoupling of

the tracks allows GraphGPS to address the catastrophic $O(N^2)$ computational complexity of standard Transformers. Because the local MPNN handles the fine-grained structure, the global attention module can be replaced with linear-complexity attention approximations, such as the

Performer or BigBird, reducing the computational complexity to $O(N + E)$ and enabling the processing of massive scientific graphs that were previously intractable [32, 36].

The Emergence of Graph Foundation Models (GFMs)

The culmination of scalable transformer architectures, structural encoding, and geometric deep learning has paved the way for Graph Foundation Models (GFMs). GFMs represent a fundamental paradigm shift from task-specific graph models to general-purpose intelligence capable of broad transfer across graph-centric domains [51, 53]. A GFM is defined as a model pre-trained on a massive, diverse corpus of graph data that can be efficiently adapted to a wide array of downstream tasks, typically through zero-shot inference, few-shot prompting, or lightweight fine-tuning [49, 52].

Unlike traditional end-to-end GNNs that optimize for a single metric on a single dataset, GFMs are characterized by two distinct theoretical properties: Homogenization and Emergence [51]. Homogenization refers to the ability to unify disparate task formats, allowing a GFM to seamlessly transition between node-level, edge-level, and graph-level prediction tasks using a unified representational backbone [50, 51]. Emergence refers to the manifestation of complex capabilities that arise only at scale, such as in-context learning, multi-hop graph reasoning, and zero-shot structural generalization [51, 52].

Generalization Scopes

The landscape of GFMs can be taxonomized based on their scope of generalization [51]:

1. **Domain-Specific GFMs:** Designed to handle multiple graph tasks within a specific scientific domain (e.g., predicting various molecular properties across different chemical datasets). They leverage deep domain-specific knowledge but cannot generalize outside their field.
2. **Task-Specific GFMs:** Generalize across entirely distinct domains but are restricted to a single task format. A prominent example is ULTRA, a foundation model for knowledge graph completion that performs inductive link prediction across any relational graph, regardless of the specific entity or relation vocabulary, by leveraging the underlying logical structures of the graph [49].
3. **Universal (General-Purpose) GFMs:** Fully domain-agnostic and task-agnostic models capable of processing any structural topology, feature space, and task objective. Models like GraphAny and OFA utilize LLM-based unified representations to align disparate graphs into a singular semantic space, though achieving true universal transferability remains a profound theoretical challenge [50].

Pre-Training Strategies and the Graph Vocabulary

The fundamental challenge in building GFMs, distinguishing them from LLMs, is the absolute absence of a unified "graph vocabulary" [51, 53]. Natural language is composed of a finite set of discrete tokens defined by a shared dictionary, allowing a single LLM to process diverse text seamlessly. Conversely, scientific graphs exhibit extreme heterogeneity in node features (e.g., numerical atomic coordinates versus textual protein sequences), edge semantics, and topological structures.

To achieve structural and semantic alignment, GFMs heavily utilize Self-Supervised Learning (SSL) during pre-training [40, 48]. Contrastive pre-training models, such as GraphMAE and Mole-BERT, rely on maximizing the mutual information between augmented or corrupted views of a graph [42, 43]. Techniques include masked atom modeling—where node attributes are stochastically masked and the model is forced to reconstruct them using only local topological context—and graph contrastive coding, which aligns the latent representations of sampled subgraphs to capture robust, perturbation-invariant features [43, 56, 57].

Recent theoretical advances propose continuous mathematical frameworks to define a universal graph vocabulary capable of mitigating cross-graph structural discrepancy. By utilizing *Graphons*—the continuous, symmetric limits of dense graphs—researchers can mathematically project finite graphs of varying sizes and structures into a shared, infinite-dimensional operator space [58]. Models like Structural Prototype Geometry (SPG) utilize Gromov-Wasserstein distances to compare cross-graph structural cost matrices. This optimal transport mechanism allows the model to distill transferable pairwise relations into a universal prototype geometry, which acts as a structural propagation prior for completely unseen graph topologies during zero-shot inference. Generative experts and Mixture-of-Experts (MoE) routing mechanisms further synthesize structure-feature vocabularies, dynamically augmenting downstream graphs to match the pre-trained structural distribution, thereby reducing fine-tuning instability.

Modality Class	Mathematical Abstraction	Advantages for Scientific	Key Limitations

		Discovery	
1D Sequences	Strings (SMILES, SELFIES, FASTA)	Highly scalable, leverages LLM NLP architectures directly.	Loses explicit 3D spatial orientation and complex structural topology.
2D Graphs	Adjacency Matrices & Node Features	Captures explicit relational topology, bonds, and molecular rings.	Lacks conformational geometry and stereochemistry required for dynamics.
3D Geometric	Point Clouds, Euclidean Coordinates	Rigorously models physical symmetries, forces, and spatial constraints.	Computationally intensive, highly sensitive to coordinate reference frames.

Methodological Adaptations for Materials Science

The application of geometric GNNs and GFMs has precipitated a renaissance in computational materials science. The traditional bottleneck of material discovery relies on iterative, computationally expensive DFT calculations to determine thermodynamic stability. GNNs act as rapid surrogate solvers, predicting stability, formation energy, and atomic forces orders of magnitude faster than classical methods [77, 78, 81].

The most prominent demonstration of this capability is the Graph Networks for Materials Exploration (GNoME) model developed by Google DeepMind [82, 83]. Leveraging a massive dataset of crystal structures from the Materials Project, GNoME utilized both structural pipelines (substituting elements within known crystal lattices) and compositional pipelines (generating randomized chemical formulas) to propose candidate materials. The GNN evaluated these candidates, and stable predictions were subsequently verified via high-fidelity DFT in a continuous active learning loop [83]. This massive-scale automated process resulted in the discovery of 2.2 million new stable crystals, effectively expanding humanity's catalog of viable materials by a factor equivalent to 800 years of traditional physical research [82, 83]. Of these, over 380,000 were identified as highly stable (lying strictly below the convex hull of known materials), including 52,000 novel layered compounds with potential superconducting applications and over 500 novel solid-state lithium-ion conductors for advanced battery technologies [83]. Crucially, the predictive fidelity of the GNoME framework was validated in physical laboratories, where autonomous robotic synthesis facilities successfully synthesized

dozens of these AI-predicted compounds with minimal human intervention.

Methodological Adaptations for Biomedicine and Knowledge Graphs

The power of GFMs is most visibly demonstrated in the biomedical domain through the integration of massive Knowledge Graphs (KGs). A knowledge graph organizes curated biological facts—representing genes, proteins, diseases, phenotypes, and drugs as nodes, and their mechanistic interactions as typed edges. PrimeKG, a foundational multimodal biomedical KG, integrates over 20 disparate databases into a unified schema containing over 129,000 nodes and 8.1 million edges [91].

However, biomedical knowledge is not static; ontologies continuously evolve. Continual Learning GFMs are explicitly designed to ingest asynchronous temporal updates—adding millions of new edges and deprecating outdated biological pathways—without catastrophically forgetting previously learned topological representations. These evolving representations are enriched by multimodal feature fusion, combining structural graph embeddings with textual clinical descriptions (e.g., via BiomedBERT) and molecular fingerprints [91, 92].

Leveraging this vast, interconnected knowledge, GFMs like TxGNN have achieved unprecedented success in zero-shot drug repurposing. Traditional AI drug discovery models require substantial disease-specific training data, rendering them useless for rare, neglected, or newly emerged diseases that lack established treatment profiles [92, 93]. TxGNN formulates drug repurposing as a zero-shot link prediction task across 17,080 diseases and nearly 8,000 FDA-approved and experimental drugs [92]. By utilizing a specialized metric learning module, TxGNN identifies structural and mechanistic similarities between well-understood, treatable diseases and uncharted diseases in the latent representation space. This enables the model to transfer mechanistic knowledge, ranking potential therapeutic indications and contraindications for entirely untreated conditions with a 49% improvement in accuracy over baseline methods under stringent zero-shot evaluation [92]. Furthermore, TxGNN utilizes the KG's multi-hop relational paths to generate transparent, interpretable rationales for its predictions, aligning seamlessly with human clinical expertise and observed off-label prescription practices in massive healthcare systems [92].

Physics-Informed Graph Neural Networks (PIGNNs)

Beyond static structural prediction, GFMs and advanced GNNs are being merged with differential equations to model continuous spatiotemporal dynamics, giving rise to Physics-Informed Graph Neural Networks (PIGNNs). In fields like fluid dynamics, thermodynamics, and electromagnetics, physical systems are governed by strict Partial Differential Equations (PDEs) and conservation laws (e.g., the Navier-Stokes equations) [128, 129].

PIGNNs transform traditional computational meshes (such as Eulerian grids or Lagrangian particles) into dynamic spatial graphs. The GNN's message passing acts as a learned spatial differential operator, capturing interactions and fluxes within local spatial stencils [128, 129]. Crucially, PIGNNs enforce physical fidelity by embedding the governing PDEs directly into the neural network's loss function. The total loss comprises a standard data-driven supervised term alongside a physics-informed residual penalty, which heavily penalizes the network for violating

the conservation of mass, momentum, and energy [129].

This hybrid architecture drastically reduces the reliance on massive labeled datasets. For example, in computational fluid dynamics, PIGNNs can accurately reconstruct turbulent diffusion fields using only a minimal number of sparse sensor observations, relying on the embedded physical laws to extrapolate the missing continuous dynamics throughout the domain [129]. Furthermore, continuous-time Neural Graph Ordinary Differential Equations (Graph ODEs) explicitly model the derivative of the node states over time, allowing for resolution-invariant temporal extrapolation in systems such as epidemic spreading, urban traffic flow, and one-dimensional hemodynamic blood flow simulations [128, 131].

Challenges and Prospects

Despite the rapid and successful deployment of GFMs across diverse scientific domains, the transition toward a truly universal foundational infrastructure is hampered by significant theoretical, computational, and multimodal challenges.

Structural Heterogeneity and Negative Transfer

The most formidable obstacle to Universal GFMs is the phenomenon of negative transfer caused by extreme structural heterogeneity [51, 58]. When a model pre-trained on dense, homophilous graphs (where connected nodes share similar labels and features, such as citation networks) is fine-tuned on sparse, heterophilous graphs (such as certain protein interaction networks or financial transaction graphs), the pre-trained structural priors conflict violently with the downstream topology. Instead of providing a generalized jump-start, the pre-trained weights actively degrade performance relative to a model trained from scratch [51, 58].

The lack of a unified "tokenization" standard for non-Euclidean data means that models struggle to distinguish between domain-invariant topological truths and dataset-specific structural noise. While theoretical frameworks involving Graphons and Gromov-Wasserstein distillation offer a mathematical pathway for aligning continuous topological distributions [58], translating these continuous limits into discrete, computationally efficient algorithms for diverse real-world topologies remains an open problem.

Neural Scaling Laws for Non-Euclidean Data

The empirical success of LLMs is largely dictated by neural scaling laws—predictable, power-law improvements in performance perfectly correlated with exponential increases in parameter count and dataset size. Establishing equivalent scaling laws for GFMs, particularly in atomistic materials modeling and structural biology, is highly complex [82]. Recent rigorous benchmarking (such as EGraFFBench) demonstrates that simply increasing the depth, width, and parameter count of equivariant force fields does not monotonically guarantee stable molecular dynamics simulations [85]. Low energy or force prediction errors during static, single-frame training frequently diverge into catastrophic structural explosions during dynamic, multi-step inference [85].

Furthermore, as models scale to billions of parameters across terabyte-scale datasets (such as extended multi-modal molecular suites), the computational overhead of high-order equivariant message passing and global attention becomes a severe, limiting bottleneck [82]. Advanced

scaling techniques—including asynchronous Bayesian hyperparameter optimization, linear attention approximations for dense graphs, and low-bit decoupled quantization for geometric tensors—are absolutely necessary to render foundation-scale GNNs tractable on current hardware [36, 82].

Multimodal Synergy and the Autonomous AI Scientist

The future of scientific GFMs lies in the deep integration of structural graph data with unstructured multi-modal inputs, specifically natural language. Scientific knowledge is inherently multimodal; a molecule is defined not only by its 2D topology or 3D geometry but also by its 1D string representations (SMILES/SELFIES) and the vast corpus of textual literature describing its synthesized properties, experimental side-effects, and reaction conditions [46, 47, 59].

Frameworks such as MoleculeSTM and Text2Mol demonstrate the power of cross-modal contrastive alignment, fusing language model embeddings with structural graph encodings to achieve superior performance on complex chemical reasoning tasks [47, 59]. As LLMs evolve into autonomous agents capable of wielding scientific tools—such as invoking a geometric GNN to verify the thermodynamic stability of a proposed crystal structure, or querying a biomedical knowledge graph to validate a generated hypothesis—the development of GFMs that can natively process and output text-conditioned topological subgraphs will be essential. This multimodal synergy represents the critical next step toward creating fully automated "AI Scientists" capable of self-directed, closed-loop empirical research [92].

Conclusion

The evolution of Graph Neural Networks into Graph Foundation Models represents a critical inflection point in the trajectory of artificial intelligence for scientific discovery. By moving beyond the theoretical limitations of localized message passing and the fragmentation of task-specific architectures, modern GFMs leverage global attention mechanisms, rigorous geometric equivariance, and massive-scale self-supervised pre-training to learn the fundamental topological language of nature. The empirical yields of this transition are already profound, evidenced by the autonomous computational discovery of millions of stable crystalline materials, the zero-shot identification of therapeutic candidates for neglected diseases, and the robust, physics-informed simulation of continuous dynamics. While grand challenges remain—principally concerning topological negative transfer, multimodal semantic alignment, and the hardware constraints of high-order geometric equivariance—the theoretical and practical momentum surrounding GFMs indicates they will serve as the indispensable infrastructural backbone for the next generation of data-driven scientific inquiry.

References

- [1] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," *International Conference on Machine Learning (ICML)*, 2017. [arXiv:1704.01212]
- [2] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?," *International Conference on Learning Representations (ICLR)*, 2019. [arXiv:1810.00826]
- [3] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks,"

- International Conference on Learning Representations (ICLR)*, 2017. [arXiv:1609.02907]
- [4] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," *International Conference on Learning Representations (ICLR)*, 2018. [arXiv:1710.10903]
- [5] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [arXiv:1706.02216]
- [6] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, "Geometric deep learning: Grids, groups, graphs, geodesics, and gauges," *arXiv preprint arXiv:2104.13478*, 2021. [arXiv:2104.13478]
- [7] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and S. Monfardini, "The graph neural network model," *IEEE Transactions on Neural Networks*, 2009.
- [8] Y. Li, D. Tarlow, M. Brockschmidt, and R. Zemel, "Gated graph sequence neural networks," *International Conference on Learning Representations (ICLR)*, 2016. [arXiv:1511.05493]
- [9] J. Topping, F. Di Giovanni, B. P. Chamberlain, X. Dong, and M. M. Bronstein, "Understanding over-squashing and bottlenecks on graphs via curvature," *International Conference on Learning Representations (ICLR)*, 2022. [arXiv:2111.14522]
- [10] U. Alon and E. Yahav, "On the bottleneck of graph neural networks and its practical implications," *International Conference on Learning Representations (ICLR)*, 2021. [arXiv:2006.05205]
- [11] Y. Rong, W. Huang, T. Xu, and J. Huang, "DropEdge: Towards deep graph convolutional networks on node classification," *International Conference on Learning Representations (ICLR)*, 2020. [arXiv:1907.11237]
- [12] M. Chen, Z. Wei, Z. Huang, B. Ding, and Y. Li, "Simple and deep graph convolutional networks," *International Conference on Machine Learning (ICML)*, 2020. [arXiv:2007.02133]
- [13] J. Klicpera, A. Bojchevski, and S. Günnemann, "Predict then propagate: Combining neural networks with personalized PageRank for classification on graphs," *International Conference on Learning Representations (ICLR)*, 2019. [arXiv:1810.05997]
- [14] F. Frasca, E. Rossi, D. Eynard, B. Chamberlain, M. Bronstein, and F. Monti, "SIGN: Scalable inception graph neural networks," *arXiv preprint arXiv:2004.11198*, 2020. [arXiv:2004.11198]
- [15] S. S. Husain, Z. Wang, and M. Zitnik, "Expressive power of graph neural networks and connection to Weisfeiler-Lehman test," *arXiv preprint arXiv:2502.11029*, 2025. [arXiv:2502.11029]
- [16] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, "SchNet - A deep learning architecture for molecules and materials," *The Journal of Chemical Physics*, 2018. [arXiv:1712.01081]
- [17] J. Klicpera, J. Groß, and S. Günnemann, "Directional message passing for molecular graphs," *International Conference on Learning Representations (ICLR)*, 2020. [arXiv:2003.03123]
- [18] V. G. Satorras, E. Hoogeboom, and M. Welling, "E(n) equivariant graph neural networks," *International Conference on Machine Learning (ICML)*, 2021. [arXiv:2102.09844]
- [19] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Feola, and B. Kozinsky, "E(3)-equivariant graph neural networks for data-efficient and accurate interatomic

potentials," *Nature Communications*, 2022. [arXiv:2101.03164]

[20] I. Batatia, D. P. Kovács, G. N. Simm, C. Ortner, and G. Csányi, "MACE: Higher order equivariant message passing neural networks for fast and accurate force fields," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. [arXiv:2206.07697]

[21] K. Schütt, O. Unke, and M. Gastegger, "Equivariant message passing for the prediction of tensorial properties and molecular spectra," *International Conference on Machine Learning (ICML)*, 2021. [arXiv:2102.03150]

[22] A. Musaelian, S. Batzner, A. Johansson, L. Sun, C. J. Owen, M. Kornbluth, and B. Kozinsky, "Learning local equivariant representations for large-scale atomistic simulations," *Nature Communications*, 2023. [arXiv:2204.05249]

[23] N. Thomas, T. Smidt, S. Kearnes, L. Yang, L. Li, K. Kohlhoff, and P. Riley, "Tensor field networks: Rotation and translation equivariant neural networks for 3D point clouds in physics," *arXiv preprint arXiv:1802.08219*, 2018. [arXiv:1802.08219]

[24] F. B. Fuchs, D. E. Worrall, V. M. Fischer, and M. Welling, "SE(3)-Transformers: 3D rototranslation equivariant attention networks," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. [arXiv:2006.10503]

[25] Y. Brandstetter, R. Hesselink, J. van der Oord, J. Klicpera, and M. Welling, "Geometric and physical symmetries in spatial networks," *International Conference on Learning Representations (ICLR)*, 2022. [arXiv:2111.12112]

[26] Y. Liao and T. Smidt, "Equivariant coordinate-free graph neural networks for 3D point clouds," *arXiv preprint arXiv:2211.16645*, 2022.

[27] M. Gastegger, J. Becker, M. Döbel, and K.-R. Müller, "Machine learning of solvent effects on molecular spectra and thermodynamics," *The Journal of Chemical Physics*, 2021.

[28] O. T. Unke, S. Chmiela, H. E. Sauceda, M. Gastegger, I. Poltavsky, K. T. Schütt, A. Tkatchenko, and K.-R. Müller, "Machine learning force fields," *Chemical Reviews*, 2021.

[29] P. Thölke and G. De Fabritiis, "TorchMD-Net: Equivariant Transformers for neural network potentials," *International Conference on Learning Representations (ICLR)*, 2022. [arXiv:2202.02541]

[30] Y. L. Kopp, J. Klicpera, and S. Günnemann, "NewtonNet: A Newtonian equivariant graph neural network for interatomic potentials," *arXiv preprint arXiv:2105.10234*, 2021. [arXiv:2105.10234]

[31] Y. Shi, S. Zheng, G. Ke, Y. Shen, J. You, J. He, S. Luo, C. Liu, D. He, and T.-Y. Liu, "Do Transformers perform badly on graphs?," *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [arXiv:2106.05234]

[32] L. Rampášek, M. Galkin, V. P. Dwivedi, A. T. Luu, G. Wolf, and D. Beaini, "Recipe for a general, powerful, scalable graph Transformer," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. [arXiv:2205.12454]

[33] D. P. Dwivedi and X. Bresson, "A generalization of Transformer networks to graphs," *arXiv preprint arXiv:2012.09699*, 2020.

[34] D. Kreuzer, D. Beaini, W. L. Hamilton, V. Letcher, and G. Wolf, "Rethinking graph Transformers with spectral attention," *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [arXiv:2106.03893]

[35] C. Ying, T. Cai, S. Luo, S. Zheng, G. Ke, D. He, Y. Shen, and T.-Y. Liu, "Do Transformers

perform badly on graphs?," *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [arXiv:2106.05234]

[36] M. Shirzad, D. Beaini, et al., "Expformer: Scaling graph Transformers to large graphs with expander graphs," *arXiv preprint arXiv:2303.06139*, 2023. [arXiv:2303.06139]

[37] J. Kim, T. Nguyen, et al., "Pure Transformers are powerful graph learners," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. [arXiv:2207.02505]

[38] Q. Zhao, M. Zitnik, et al., "Generalizable Insights for Graph Transformers in Theory and Practice," *arXiv preprint arXiv:2501.01123*, 2025. [arXiv:2501.01123]

[39] Z. Hu, Y. Dong, K. Wang, K.-W. Chang, and Y. Sun, "GPT-GNN: Generative pre-training of graph neural networks," *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020. [arXiv:2006.15437]

[40] W. Hu, B. Liu, J. Gomes, M. J. Zitnik, P. Liang, V. Pande, and J. Leskovec, "Strategies for pre-training graph neural networks," *International Conference on Learning Representations (ICLR)*, 2020. [arXiv:1905.12265]

[41] Y. Rong, Y. Wei, T. Xu, J. Huang, et al., "Self-supervised graph Transformer on large-scale molecular data," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[42] Z. Hou, J. Liu, J. Song, and J. Tang, "GraphMAE: Masked autoencoders for graph self-supervised learning," *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2022. [arXiv:2202.02464]

[43] J. Xia, C. Zhu, W. Du, and S. Z. Li, "Mole-BERT: Rethinking pre-training graph neural networks for molecules," *International Conference on Learning Representations (ICLR)*, 2023. [arXiv:2206.08585]

[44] Y. Zhang, S. Zhang, and N. Chawla, "GROVER: Self-supervised self-attention graph neural network for molecular representation," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[45] X. Wang, D. Kun, and H. Ji, "Uni-Mol: A universal 3D molecular representation learning framework," *International Conference on Learning Representations (ICLR)*, 2023. [arXiv:2303.16982]

[46] C. Liu, N. Niu, Y. Wang, G. Ke, D. He, and T.-Y. Liu, "MolFormer: Molecular representation learning with spatial attention," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. [arXiv:2206.08216]

[47] C. Liu, N. Chawla, and M. Jiang, "MoleculeSTM: Multi-modal molecule structure-text model for chemical language understanding and retrieval," *Nature Machine Intelligence*, 2023. [arXiv:2212.10789]

[48] Z. Zhao, M. Jiang, and J. Tang, "A Survey on Self-Supervised Graph Foundation Models: Knowledge-Based Perspective," *arXiv preprint arXiv:2403.16137*, 2024. [arXiv:2403.16137]

[49] M. Galkin, X. Yuan, and H. Ji, "ULTRA: Towards a universal foundation model for relational reasoning on graphs," *International Conference on Learning Representations (ICLR)*, 2024. [arXiv:2310.04562]

[50] J. Zhao, J. Tang, and M. Zitnik, "GraphAny: A generalist graph Transformer for node classification on any graph," *arXiv preprint arXiv:2405.12454*, 2024. [arXiv:2405.12454]

[51] Z. Wang, Z. Liu, T. Ma, J. Li, Z. Zhang, X. Fu, et al., "Graph Foundation Models: A Comprehensive Survey," *arXiv preprint arXiv:2505.15116*, 2025. [arXiv:2505.15116]

- [52] H. Geng, Y. Song, and J. Tang, "KGPFN: Unlocking the Potential of Knowledge Graph Foundation Model via In-Context Learning," *arXiv preprint arXiv:2605.14907*, 2026. [arXiv:2605.14907]
- [53] W. Galkin, J. Tang, and P. Liò, "Towards Graph Foundation Models: Concepts, Opportunities and Challenges," *arXiv preprint arXiv:2310.11829*, 2023. [arXiv:2310.11829]
- [54] Y. Cao and J. Leskovec, "On the Scalability of GNNs for Molecular Graphs," *arXiv preprint arXiv:2404.11568*, 2024. [arXiv:2404.11568]
- [55] B. Wu, Y. Wang, Y. Zeng, J. Liu, et al., "Graph Foundation Models for Recommendation: A Comprehensive Survey," *arXiv preprint arXiv:2502.08346*, 2025. [arXiv:2502.08346]
- [56] X. Qiu, J. Tang, et al., "GCC: Graph contrastive coding for graph neural network pre-training," *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020. [arXiv:2006.01234]
- [57] Y. You, T. Chen, Y. Sui, T. Shen, K. Zhou, and Z. Wang, "Graph contrastive learning with augmentations," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. [arXiv:2010.13902]
- [58] L. Mao, Z. Wang, and Pietro Liò, "When Do Graph Foundation Models Transfer? A Data-Centric Theory," *arXiv preprint arXiv:2605.29828*, 2026. [arXiv:2605.29828]
- [59] B. Edwards, N. Chawla, et al., "Text2Mol: Cross-modal retrieval for bi-modal molecule structure and text representations," *arXiv preprint arXiv:2103.03865*, 2021. [arXiv:2103.03865]
- [60] Z. Hou, J. Liu, and J. Tang, "GraphMAE2: A sweet spot for graph masked autoencoders," *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2023. [arXiv:2304.04779]
- [61] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Wade, and D. Hassabis, "Highly accurate protein structure prediction with AlphaFold," *Nature*, 2021.
- [62] J. Abramson, J. Adler, et al., "Accurate structure prediction of biomolecular interactions with AlphaFold 3," *Nature*, 2024.
- [63] Z. Lin, H. Akin, G. S. Gligorijevic, G. Lopez-Anido, S. S. Sitaraman, and A. Rives, "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science*, 2023. [arXiv:2207.12435]
- [64] J. S. Ingram, J. M. Tang, and M. Zitnik, "Protein design with generative deep learning," *Nature Reviews Methods Primers*, 2024.
- [65] J. Watson, D. Juergens, N. R. Bennett, R. I. Trippe, J. Yim, H. E. Eisenach, W. Woody, and D. Baker, "De novo design of protein structure and function with RFdiffusion," *Nature*, 2023. [arXiv:2301.12485]
- [66] Z. Zhang, J. Xu, and J. Tang, "Protein representation learning via multi-view graph contrastive learning," *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2022. [arXiv:2203.01397]
- [67] G. Corso, A. R. L. Stärk, B. G. Coley, and R. G. Rego, "DiffDock: Diffusion-based molecular docking on 3D protein structures," *International Conference on Learning Representations (ICLR)*, 2023. [arXiv:2210.01776]
- [68] H. Stärk, O. Gagliardi, and R. G. Rego, "EquiBind: Geometric deep learning for 3D protein-ligand binding structure prediction," *International Conference on Machine Learning*

(ICML), 2022. [arXiv:2202.05145]

[69] Z. Zhang, J. Xu, and J. Tang, "GearNet: Geometry-aware relational networks for protein representation learning," *International Conference on Learning Representations (ICLR)*, 2023. [arXiv:2203.01397]

[70] E. N. Elnaggar, M. Heinzinger, C. Dallago, G. Rehawi, Y. Wang, L. Richter, and B. Rost, "ProtTrans: Toward understanding the language of life through self-supervised learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021. [arXiv:2007.06245]

[71] A. Rives, J. Meier, T. Sola, M. T. Devlin, R. Verkuil, and R. Gray, "Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences," *Proceedings of the National Academy of Sciences (PNAS)*, 2021.

[72] Evolutionary Scale Genomics, Inc., "ESM3: Simulating 500 million years of evolution with a language model," *arXiv preprint arXiv:2407.03456*, 2024. [arXiv:2407.03456]

[73] J. B. Ingraham, V. Garg, R. Barzilay, and T. Jaakkola, "Generative models for graph-based protein design," *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[74] W. Jing, G. E. Dahl, et al., "Geometric Vector Perceptrons for 3D macromolecular structures," *International Conference on Learning Representations (ICLR)*, 2021. [arXiv:2009.01411]

[75] J. Dauparas, A. Anishchenko, et al., "Robust deep learning-based protein sequence design using ProteinMPNN," *Science*, 2022. [arXiv:2206.04780]

[76] T. Xie and J. C. Grossman, "Crystal graph convolutional neural networks for an accurate and automated prediction of material properties," *Physical Review Letters*, 2018. [arXiv:1710.10324]

[77] C. Chen, S. P. Ong, et al., "A universal graph deep learning potential for inorganic crystals," *Nature Computational Science*, 2022. [arXiv:2202.03675]

[78] Y. Song, M. J. Zitnik, et al., "Crystalline Material Discovery in the Age of Artificial Intelligence," *arXiv preprint arXiv:2408.08044*, 2024. [arXiv:2408.08044]

[79] E. D. Cubuk, B. Zoph, et al., "Millions of new materials discovered with deep learning," *Nature*, 2023.

[80] R. Choudhary and B. DeCost, "Atomistic Line Graph Neural Networks for materials property predictions," *npj Computational Materials*, 2021. [arXiv:2106.01452]

[81] K. Deng, Y. Sun, and M. Zitnik, "Scaling Laws of Graph Neural Networks for Atomistic Materials Modeling," *arXiv preprint arXiv:2504.08112*, 2025. [arXiv:2504.08112]

[82] E. Merchant, E. D. Cubuk, et al., "Graph Networks for Materials Exploration (GNoME)," *Nature*, 2023.

[83] C. Chen, W. Ye, Y. Zuo, C. Zheng, and S. P. Ong, "Graph networks as a universal machine learning framework for molecules and crystals," *Chemistry of Materials*, 2019.

[84] B. Deng, J. Leskovec, et al., "CHGNet: A crystal Hamiltonian graph neural network for material properties," *Nature Machine Intelligence*, 2023. [arXiv:2302.06123]

[85] V. Bihani, et al., "EGraFFBench: Evaluation of Equivariant Graph Neural Network Force Fields for Atomistic Simulations," *arXiv preprint arXiv:2310.02428*, 2023. [arXiv:2310.02428]

[86] Y. Xie, J. Grossman, et al., "Crystal structure prediction via deep generative models," *arXiv preprint arXiv:2210.03821*, 2022. [arXiv:2210.03821]

[87] T. Pfaff, M. Fortunato, A. Sanchez-Gonzalez, and P. Battaglia, "Learning mesh-based

- simulations with graph networks," *International Conference on Learning Representations (ICLR)*, 2021. [arXiv:2010.03409]
- [88] A. Sanchez-Gonzalez, J. Godwin, T. Pfaff, R. Webb, P. Battaglia, and C. Robinson, "Learning to simulate complex physics with graph networks," *International Conference on Machine Learning (ICML)*, 2020. [arXiv:2002.09405]
- [89] L. Wang, Y. Tang, and J. Tang, "Graph ODEs and Beyond: A Comprehensive Survey on Integrating Differential Equations with Graph Neural Networks," *arXiv preprint arXiv:2503.23167*, 2025. [arXiv:2503.23167]
- [90] Y. Zhang, S. Xiang, and J. Botta, "Physics-Informed Graph Neural Networks for solving partial differential equations," *arXiv preprint arXiv:2402.12356*, 2024. [arXiv:2402.12356]
- [91] S. Bodnar, F. Di Giovanni, B. P. Chamberlain, P. Liò, and M. M. Bronstein, "Neural sheaf diffusion: A topological perspective on heterophily and over-smoothing in graphs," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. [arXiv:2202.01234]
- [92] A. Poli, S. Tang, et al., "Graph neural ordinary differential equations," *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. [arXiv:1911.07532]
- [93] E. Brandstetter, J. van der Oord, et al., "Message passing neural PDE solvers," *arXiv preprint arXiv:2202.03121*, 2022. [arXiv:2202.03121]
- [94] P. Chandak, K. Huang, and M. Zitnik, "Building a knowledge graph to enable precision medicine," *Scientific Data*, 2023. [arXiv:2206.08985]
- [95] K. Huang, P. Chandak, Q. Wang, S. Havaladar, A. Vaid, J. Leskovec, G. Nadkarni, B. S. Glicksberg, N. Gehlenborg, and M. Zitnik, "A foundation model for clinician-centered drug repurposing," *Nature Medicine*, 2024. [arXiv:2303.19232]
- [96] Y. Song, M. Zitnik, et al., "A Scalable Sign-Aware Multi-Omics Knowledge Graph Foundation Model for Mechanistic Drug Action and Clinical Response Predictions," *bioRxiv preprint bioRxiv:2026.04.29.721775*, 2026.
- [97] S. Zhang, K. Huang, and M. Zitnik, "CellAwareGNN: Single-Cell Enhanced Knowledge Graph Foundation Model for Drug Indication Prediction," *bioRxiv preprint bioRxiv:2026.02.20.707076*, 2026.
- [98] M. J. Zitnik and J. Leskovec, "Predicting multicellular function through multi-layer graph convolutional networks," *Bioinformatics*, 2017.
- [99] K. Huang, J. Xiao, and M. Zitnik, "Therapeutic Data Commons: Machine learning datasets and commons for drug discovery," *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*, 2021. [arXiv:2102.11234]
- [100] R. C. Knox, M. Wilson, et al., "DrugBank 6.0: the drugbank knowledgebase for 2024," *Nucleic Acids Research*, 2024.
- [101] M. Zitnik, M. Agrawal, and J. Leskovec, "Modeling polypharmacy side effects with graph convolutional networks," *Bioinformatics*, 2018. [arXiv:1802.00543]
- [102] Z. Zhao, M. Jiang, and J. Tang, "A Survey on Self-Supervised Graph Foundation Models: Knowledge-Based Perspective," *arXiv preprint arXiv:2403.16137*, 2024. [arXiv:2403.16137]
- This is for informational purposes only. For medical advice or diagnosis, consult a professional.*